

overlap-calculator

User Manual

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overlap-calculator is a Python package for reproducible batch analysis of the spectral overlap between a molecular absorption spectrum and a reference light source. It treats two input families as equal first-class citizens in the same run:

- **Theoretical** — Gaussian TD-DFT `.out` / `.log` files.
- **Experimental** — tabular spectra as **CSV** (`.csv`), **Excel workbook** (`.xlsx`), or **legacy Excel** (`.xls`), with multi-sheet and multi-series selection.

Every TD-DFT spectrum is broadened with both Gaussian and Lorentzian line shapes, and the tool reports per-light-source absorbed-flux, absorbed-fraction, and shape-overlap metrics together with publication-grade TIFF plots.

This manual covers installation, the CLI, the HTTP API, the input and output schemas, the physical assumptions and exact formulas, and environment configuration. For worked examples on real data, see the `case_studies/` directory of the repository.

Project home: <https://github.com/pinarsyt/overlap-calculator>

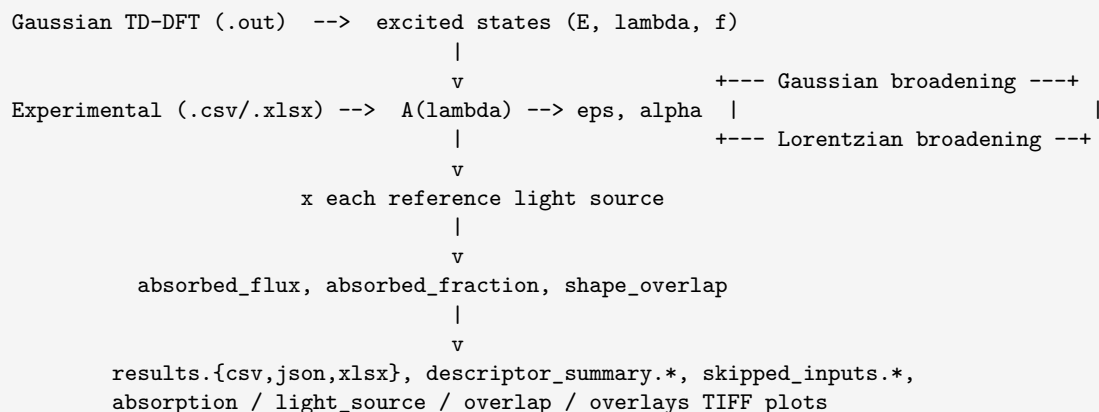
Archived release (all versions): <https://doi.org/10.5281/zenodo.19944515>

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Contents

1	Overview and Workflow	3
2	Installation	3
3	Quick Start	4
4	Command-Line Interface	5
4.1	<code>generate-input</code>	5
4.2	<code>analyze</code>	5
4.3	<code>version</code>	6
5	HTTP API	6
5.1	Endpoints	6
5.2	Example	7
6	Input Format	7
6.1	Accepted input files	7
6.2	Manifest (<code>input.json</code>)	7
6.3	Experimental data template	8
6.4	Custom light sources	9
7	Output Schema	9
7.1	<code>results.* columns</code>	10
7.2	<code>descriptor_summary.*</code>	10
7.3	<code>ranking_by_light_source__<metric>.*</code> and <code>ranking_by_sample__<metric>.*</code>	10
7.4	<code>skipped_inputs.*</code>	11
7.5	<code>results_timings.*</code>	11
7.6	Plots	11
8	Bundled Light Sources	12
9	Methodology	12
9.1	Extinction spectrum from TD-DFT	12
9.2	Beer–Lambert absorptance	13
9.3	Light-overlap metrics	13
9.4	Choosing between <code>absorbed_fraction</code> and <code>shape_overlap</code>	13
9.5	Skip policy	14
10	Configuration and Environment Variables	14
11	Troubleshooting	14
12	Citation and License	15

1 Overview and Workflow



Pipeline stages:

1. **Input resolution** — A JSON manifest can be supplied directly or auto-generated from a directory of raw files. Experimental inputs expand to one row per numeric series column.
2. **Parsing** — Regex-based TD-DFT parser for .out files; pandas-based wavelength / signal loader for tabular spectra.
3. **Spectrum reconstruction** — Molar extinction $\varepsilon(\lambda)$ from TD-DFT states via oscillator-strength weighting on a shared wavelength grid; experimental signals are treated as absorbance directly.
4. **Beer–Lambert** — $\alpha(\lambda) = 1 - 10^{-A(\lambda)}$ with configurable reference concentration and path length.
5. **Overlap metrics** — Per (sample, light source, broadening) triplet.
6. **Export** — Tabular (CSV / JSON / XLSX), descriptor summary, skip report, and TIFF plots.

2 Installation

Requirements

- Python **3.12.x**
- Optional: Conda (recommended) or Docker

Conda (recommended)

```

conda env create -f environment.yml
conda activate overlap-calculator
pip install -e .
  
```

venv / pip

```

python -m venv .venv
source .venv/bin/activate          # Linux / macOS
  
```

```
# Windows PowerShell: .venv\Scripts\Activate.ps1
# Windows CMD:       .venv\Scripts\activate.bat
pip install -e .
```

Docker

The Docker image exposes the Flask API on port 8000 *inside* the container. To keep it from clashing with a native Flask instance on the host (which uses port 8000 directly), publish it on host port 8080:

```
docker build -t overlap-calculator .
docker run --rm -p 8080:8000 overlap-calculator
curl http://localhost:8080/health
```

Docker Compose

For a managed runtime (restart policy, healthcheck, pre-populated `GAUSS_*` environment), use the bundled compose file. It uses the same **host port 8080** mapping as `docker run` above, so the native instance on 8000 and either docker variant on 8080 can run side by side:

```
docker compose up -d          # starts the `overlap-calculator` service
curl http://localhost:8080/health
docker compose down
```

The compose service uses a Python-stdlib healthcheck against `/health` and restarts unless explicitly stopped. All runtime settings in §10 can be overridden via the `environment:` block or an adjacent `.env` file.

Verify the install

```
overlap-calculator version
overlap-calculator --help
```

3 Quick Start

Drop any combination of the supported input files into `input/files/` — TD-DFT outputs, CSV spectra, and Excel workbooks can live side by side in the same directory:

```
input/files/
|-- slurm-5473089.out      # Gaussian TD-DFT output (theoretical)
|-- dye_panel.xlsx        # Excel workbook, one or more sheets (experimental)
`-- uv_vis_run3.csv       # CSV spectrum (experimental)
```

Then run the two-step pipeline:

```
overlap-calculator generate-input --files-dir input/files --out input/input.json
overlap-calculator analyze --input input/input.json --out output
```

Results land under `output/tables/` and `output/plots/` (see §7).

4 Command-Line Interface

The package installs a single entry point, `overlap-calculator`, with three subcommands.

4.1 generate-input

```
overlap-calculator generate-input --files-dir PATH [--out PATH] [--log-level LEVEL] [--log-
↪ format FORMAT]
```

Flag	Default	Description
<code>--files-dir</code>	<code>input/files</code>	Directory containing TD-DFT and experimental files
<code>--out</code>	<code>input/input.json</code>	Output manifest path
<code>--log-level</code>	<code>INFO</code>	DEBUG, INFO, WARNING, ERROR, CRITICAL
<code>--log-format</code>	<code>text</code>	text or json

4.2 analyze

```
overlap-calculator analyze --input PATH [--out PATH] [--sigma-ev FLOAT] [--wl-min FLOAT] [--
↪ wl-max FLOAT] [--num-points INT] [--concentration-m FLOAT] [--path-cm FLOAT] [--default-
↪ light-sources STR] [--light-source-file PATH]... [--plot-outputs / --no-plot-outputs] [--
↪ plot-dpi INT] [--ranking-outputs / --no-ranking-outputs] [--log-level LEVEL] [--log-
↪ format FORMAT]
```

Flag	Default	Description
<code>--input</code>	<code>input/input.json</code>	Manifest JSON (see §6)
<code>--out</code>	<code>output</code>	Output directory (created if absent)
<code>--sigma-ev</code>	<code>0.30</code>	Broadening standard deviation, eV
<code>--wl-min</code>	<code>200.0</code>	Lower bound of the wavelength grid, nm
<code>--wl-max</code>	<code>800.0</code>	Upper bound of the wavelength grid, nm
<code>--num-points</code>	<code>10000</code>	Number of grid points
<code>--concentration-m</code>	<code>1e-5</code>	Beer-Lambert reference concentration, mol L ⁻¹
<code>--path-cm</code>	<code>1.0</code>	Beer-Lambert reference path length, cm
<code>--default-light-sources</code>	<code>AM15G,LEDB4,...</code>	Comma-separated bundled-source list
<code>--light-source-file</code>	<code>none</code>	Repeatable path to a custom light-source CSV
<code>--plot-outputs</code>	<code>on</code>	Emit TIFF plots (<code>--no-plot-outputs</code> to disable)
<code>--plot-dpi</code>	<code>400</code>	TIFF plot resolution; use 600 for larger publication exports
<code>--ranking-outputs</code>	<code>on</code>	Emit grouped ranking tables and bar charts (<code>gaussian/lorentzian × absorbed_fraction / shape_overlap</code>); disable with <code>--no-ranking-outputs</code>

Flag	Default	Description
<code>--log-level</code>	INFO	Logging verbosity
<code>--log-format</code>	text	text or json

4.3 version

Prints the installed package version.

```
overlap-calculator version
```

5 HTTP API

The examples in this section use port 8000, which is the **native Flask** port (started with `python -m overlap_calculator.api.app`). If you run the same API under **Docker** (either `docker run -p 8080:8000` or `docker compose up`), use port 8080 instead — the request bodies are otherwise identical.

5.1 Endpoints

Method	Path	Purpose
GET	/health	Liveness probe — returns <code>{"status": "ok"}</code>
POST	/analyze	Multipart upload; returns a ZIP of results

POST /analyze accepts the following form fields:

Form field	Type	Description
files	file[] (req.)	Mixed .out/.csv/.xlsx/.xls inputs
light_source_files	file[] (opt.)	Custom light-source CSVs
light_source_csv	file (opt.)	Back-compat alias for a single custom CSV
sigma_ev	float (opt.)	Broadening σ , eV (default 0.30)
wl_min, wl_max	float (opt.)	Wavelength window bounds, nm
num_points	int (opt.)	Grid size (default 10000)
concentration_m	float (opt.)	Beer–Lambert reference concentration (default 1e-5)
path_cm	float (opt.)	Beer–Lambert reference path length (default 1.0)
default_light_sources	str (opt.)	Comma-separated bundled-source list
plot_outputs	bool (opt.)	Enable / disable plot generation
plot_dpi	int (opt.)	TIFF plot resolution (default 400; use 600 for larger exports)
ranking_outputs	bool (opt.)	Enable / disable grouped ranking tables and bar charts (default true)
sheet_overrides	str (opt.)	JSON object <code>{"<original-filename>": "<sheet-name>"}</code> forcing a specific sheet for uploaded .xlsx/.xls workbooks

Response: `application/zip` containing every file `overlap-calculator analyze` would have written. Uploads are processed in a per-request workspace and are not persisted on the server.

5.2 Example

```
curl -X POST http://localhost:8000/analyze \
-F "files=@input/files/theoretical.out" \
-F "files=@input/files/experimental.xlsx" \
-F "sigma_ev=0.25" \
-F "plot_dpi=400" \
-F "ranking_outputs=true" \
-F "default_light_sources=AM15G,LEDB4" \
--output analysis_outputs.zip
```

To force a specific sheet in a multi-sheet workbook, add a `sheet_overrides` form field whose value is a JSON object keyed by the original filename:

```
curl -X POST http://localhost:8000/analyze \
-F "files=@input/files/dye_panel.xlsx" \
-F 'sheet_overrides={"dye_panel.xlsx":"raw"}' \
--output analysis_outputs.zip
```

Filenames that do not appear in the map use the same auto-discovery as `generate-input`: the first sheet with a wavelength column and at least one numeric signal column is selected.

A Postman collection is provided at `docs/postman_collection.json`.

6 Input Format

6.1 Accepted input files

`overlap-calculator` accepts two input families in the same run. Both are first-class: a single manifest can mix them freely, and the pipeline produces identical downstream columns, plots, and overlap metrics.

Family	Extensions	Parser	Notes
Theoretical	.out, .log	Regex-based Gaussian TD-DFT parser	File must contain a TD route and Excited State lines
Experimental	.csv	<code>pandas.read_csv</code>	One wavelength column and one or more numeric signal columns
Experimental	.xlsx, .xls	<code>pandas.read_excel</code> (openpyxl/xlrd)	Multi-sheet workbooks supported via <code>sheet_name</code> ; <code>generate-input</code> auto-selects the first analyzable sheet

Bulk uploads via the HTTP API share the same matrix (see `GAUSS_UPLOAD_ALLOWED_EXTS` in §10).

6.2 Manifest (`input.json`)

```
[
  {
    "input_type": "theoretical",
    "sample_id": "B1",
    "source_path": "files/slurm-5473089.out",
    "series_name": null,
```

```

    "sheet_name": null
  },
  {
    "input_type": "experimental",
    "sample_id": "channel_A",
    "source_path": "files/dye_panel.xlsx",
    "series_name": "channel_A",
    "sheet_name": "raw"
  }
]

```

Field	Req?	Description
input_type	yes	"theoretical" (TD-DFT .out) or "experimental" (tabular spectrum)
sample_id	yes	Unique sample identifier; used in filenames and result rows
source_path	yes	Path to the raw file; resolved relative to the manifest, then to the CWD
series_name	exp.	Column name of the numeric series in a tabular file
sheet_name	exp.	Sheet name for .xlsx/.xls inputs; in hand-written manifests, omit or null to read the first workbook sheet

generate-input fills these automatically. The `sample_id` is derived as follows:

- **Theoretical** (.out) — if the Gaussian route contains a `%chk=<name>.chk` line, the checkpoint stem is used (e.g. `%chk=B1_td.chk` $\Rightarrow$ `sample_id = "B1_td"`). This keeps human-meaningful molecule labels even when the raw files are queue-system artefacts such as `slurm-5473089.out`. If no `%chk` line is present, the file stem is used as a fallback.
- **Experimental** — one manifest entry is produced per numeric series column, with `sample_id = "<series_name>"` (the series column name itself, so `Coumarin_1` rather than a long file-stem prefix).

If a manifest is supplied by hand and `sample_id` is omitted, the same resolution rule is applied at analysis time.

6.3 Experimental data template

The same column layout applies to CSV, XLSX, and XLS files.

CSV (.csv)

```

wavelength_nm,sample_A,sample_B
300,0.12,0.08
310,0.15,0.10
320,0.18,0.14

```

Excel workbook (.xlsx / .xls). The first row is interpreted as the header. Any numeric columns beside the wavelength axis are treated as independent samples:

wavelength_nm	channel_A	channel_B	channel_C
300	0.12	0.08	0.05
310	0.15	0.10	0.07

wavelength_nm	channel_A	channel_B	channel_C
320	0.18	0.14	0.09

Multi-sheet workbooks are fully supported. During `generate-input`, if no sheet override is supplied, the workbook sheets are scanned in order and the first sheet with a wavelength column and at least one numeric signal column is selected. In a hand-written manifest, if `sheet_name` is omitted from an entry (or set to `null`), analysis reads the first workbook sheet. To force a specific sheet, edit the generated manifest entry or author it by hand:

```
{
  "input_type": "experimental",
  "sample_id": "dye_panel__channel_A",
  "source_path": "files/dye_panel.xlsx",
  "series_name": "channel_A",
  "sheet_name": "raw"
}
```

Shared requirements (both formats).

- Exactly one wavelength column; accepted aliases (case-insensitive): `wavelength`, `wavelength_nm`, `lambda`, `wl`, `nm`. If no named wavelength column is found, the first numeric column is used.
- One or more numeric signal columns. Each signal is interpreted as absorbance $A(\lambda)$ and fed directly to the Beer–Lambert absorptance definition (§9).
- Rows with non-numeric or missing values in either the wavelength or signal column are dropped silently; a series must retain at least 10 valid rows to be analysed.

6.4 Custom light sources

A custom light source is a two-column CSV with one wavelength column and a single numeric intensity column. The filename stem becomes the `light_source_name` in the results table.

7 Output Schema

```
output/
|-- tables/
|   |-- results.{csv,json,xlsx}
|   |-- results_timings.{csv,json,xlsx}
|   |-- descriptor_summary.{csv,xlsx}
|   |-- ranking_by_light_source__<metric>.{csv,json,xlsx}    # 4 metric variants
|   |-- ranking_by_sample__<metric>.{csv,json,xlsx}          # 4 metric variants
|   |-- skipped_inputs.{csv,json}                            # emitted only if any inputs were skipped
`-- plots/
    |-- absorption/<sample>__{absolute,normalized}.tiff
    |-- light_source/<light>.tiff
    |-- overlap/<sample>__<light>__combined__{absolute,normalized}.tiff
    |-- overlap/<sample>__<light>__<method>__{absolute,normalized}.tiff
    |-- overlays/absorptance_overlay__<method>__{absolute,normalized}.tiff
    |-- ranking/by_light_source/<metric>/<light>.tiff         # bar chart per light source
    |-- ranking/by_sample/<metric>/<sample>.tiff              # bar chart per sample
```

`<metric>` cycles through the four overlap descriptors: `gaussian_absorbed_fraction`, `lorentzian_absorbed_fraction`, `gaussian_shape_overlap`, `lorentzian_shape_overlap`. The

ranking tables and bar charts can be suppressed with `--no-ranking-outputs` (CLI) or `ranking_outputs=false` (API).

7.1 results.* columns

One row per (sample_id, light_source_name) pair. Shared columns:

Column	Description
source_type	theoretical or experimental
sample_id	Unique sample identifier
light_source_name	AM15G, LEDB1, LEDB2, LEDB3, LEDB4, CIEFL10, or custom stem
sigma_ev	Broadening σ used, eV (theoretical inputs)
light_flux_total	$\int I(\lambda) d\lambda$ over the working grid
reference_concentration_molar	Beer-Lambert c used, mol L ⁻¹
reference_path_cm	Beer-Lambert L used, cm
absorption_unit	Unit of ε / A in the exported row
light_source_unit	Unit of $I(\lambda)$ (e.g. W m ⁻² nm ⁻¹ for AM1.5G)
absorbed_flux_unit	Unit of the absorbed-flux integral
td_method_basis	Route metadata from the TD-DFT file (null for experimental)
td_solvent	PCM solvent from the TD-DFT file (null for experimental)

Per-broadening columns (prefixes `gaussian_` and `lorentzian_`):

Suffix	Meaning
lambda_max_nm	Peak wavelength of the broadened absorbance
molar_absorptivity_max	Peak ε in M ⁻¹ cm ⁻¹ (theoretical only)
absorbance_max	Peak Beer-Lambert $\alpha(\lambda)$
absorbed_flux	$\int \alpha(\lambda) I(\lambda) d\lambda$
absorbed_fraction	$\text{absorbed_flux} / \text{light_flux_total} \in [0, 1]$
shape_overlap	Dimensionless shape comparator on max-normalised spectra

Broadening deltas (Gaussian – Lorentzian): `delta_absorbed_flux`, `delta_absorbed_fraction`, `delta_shape_overlap`, `abs_delta_absorbed_fraction`.

7.2 descriptor_summary.*

A condensed table of the most actionable per-sample descriptors across all light sources, suitable as a starting point for ranking.

7.3 ranking_by_light_source__<metric>.* and ranking_by_sample__<metric>.*

Two complementary grouped rankings answer the two practical questions that arise from a multi-sample, multi-light-source run:

- **ranking_by_light_source__<metric>.{csv,json,xlsx}** — within each light source, samples are ordered from the highest to the lowest value of the chosen overlap metric. Use this to ask: “Given this light source, which molecule absorbs (or matches in shape) best?”

- **ranking_by_sample_<metric>.{csv,json,xlsx}** — within each sample, light sources are ordered the same way. Use this to ask: “*For this molecule, which light source gives the strongest overlap?*”

Each table is emitted four times — once per **<metric>** — covering both broadening methods (**gaussian_***, **lorentzian_***) and both metric families (**absorbed_fraction**, **shape_overlap**):

<metric>	Sorts by
gaussian_absorbed_fraction	Beer–Lambert absorbed fraction with Gaussian broadening; intensity-weighted
lorentzian_absorbed_fraction	Beer–Lambert absorbed fraction with Lorentzian broadening; intensity-weighted
gaussian_shape_overlap	Shape-only correlation (intensity-independent) under Gaussian broadening
lorentzian_shape_overlap	Shape-only correlation (intensity-independent) under Lorentzian broadening

Both interpretations are reported because they answer different scientific questions: **absorbed_fraction** tracks the fraction of incident photons that are actually absorbed under the chosen Beer–Lambert reference (c, L) and is the right figure of merit for photocatalysis, photovoltaic action, or photo-excitation yield; **shape_overlap** is a brightness-independent spectral matching score, appropriate when comparing differently-scaled illuminants on equal footing (§9.4).

Each row carries a **rank** column (1 = best within the group) and all four metric columns side by side, so the file can be re-sorted client-side without rerunning the pipeline. Companion bar charts are written to **plots/ranking/by_light_source/<metric>/<light>.tiff** and **plots/ranking/by_sample/<metric>/<sample>.tiff**; titles and axis labels embed the chosen metric and the Beer–Lambert reference conditions to keep the figures unambiguous.

7.4 skipped_inputs.*

Emitted whenever at least one input is rejected. The **tables/skipped_inputs.{csv,json}** rows carry **sample_id** and **reason** columns. The manifest-side **input/skipped_inputs.json** produced by **generate-input** additionally carries the resolved **source_path**. Each **reason** string is prefixed with the stage at which the input was skipped (**INPUT_SKIPPED** / **INPUT_ERROR** / **ANALYSIS_ERROR**) followed by the error message.

7.5 results_timings.*

One row per (**sample_id**, **light_source_name**) pair. Shared columns: **source_type**, **sample_id**, **light_source_name**, **plot_ms**. Each broadening method contributes a five-column block (**gaussian_parse_tddf**, **gaussian_broadening_ms**, **gaussian_light_source_ms**, **gaussian_integrals_ms**, **gaussian_total_sample_ms** and the matching **lorentzian_*** quintuplet). The row ends with **total_pair_ms** (sum of both methods’ **total_sample_ms**) and **total_with_plot_ms** (**total_pair_ms** + **plot_ms**).

7.6 Plots

TIFF plots are emitted at 400 dpi by default and include both absolute and max-normalised variants. Use **--plot-dpi600** on the CLI or **plot_dpi=600** in the API when larger publication-resolution files are worth the additional disk and transfer size.

Directory	Content
absorption/	Per-sample absorptance spectra, both broadenings overlaid

Directory	Content
light_source/	Each reference light source in isolation
overlap/<combined>	(sample, light) overlap — both broadenings on one axes
overlap/<method>	(sample, light) overlap — single broadening
overlays/	Multi-sample absorbance overlays per broadening
ranking/by_light_s ource/<metric>/	Bar chart per light source ranking samples by <metric> (descending)
ranking/by_sample/ <metric>/	Bar chart per sample ranking light sources by <metric> (descending)

The overlap plots shade the geometric overlap area $\min(\alpha(\lambda), \hat{I}(\lambda))$ (absolute panel) and $\min(\hat{\alpha}(\lambda), \hat{I}(\lambda))$ (normalised panel). In the normalised panel the shaded region is exactly the integrand of `shape_overlap`, so the figure and the numerical metric report the same quantity: the area bounded above by the lower of the two curves at every wavelength.

8 Bundled Light Sources

Name	Source	Unit
AM15G	NREL ASTM G-173 AM1.5G reference solar spectrum	$\text{W m}^{-2} \text{ nm}^{-1}$
LEDB1	CIE 15 illuminant LED-B1 (opt-in; not in default set)	relative (shape only)
LEDB2	CIE 15 illuminant LED-B2	relative (shape only)
LEDB3	CIE 15 illuminant LED-B3	relative
LEDB4	CIE 15 illuminant LED-B4	relative
CIEFL10	CIE 15 fluorescent illuminant FL10	relative

Unit policy: because **AM15G** carries an absolute spectral irradiance, `absorbed_flux` for AM1.5G has units of W m^{-2} ; for the relative CIE LED / FL sources `absorbed_flux_unit` is `relative_integral` and only `absorbed_fraction` and `shape_overlap` are directly comparable across dyes.

Third-party spectral data. The LEDB1, LEDB2, LEDB3, LEDB4, and CIEFL10 spectra are official CIE data sets derived from *CIE 015:2018 Colorimetry, 4th Edition* (International Commission on Illumination, <https://cie.co.at/>). The AM15G spectrum is the ASTM G-173-03 AM1.5 global reference distributed by the U.S. National Renewable Energy Laboratory (NREL). See §12 for full citations and BibTeX entries.

9 Methodology

9.1 Extinction spectrum from TD-DFT

Each excited state i contributes a normalised line profile in wavenumber space $\tilde{\nu} = 10^7/\lambda[\text{nm}]$ (cm^{-1}):

$$g_i(\tilde{\nu}) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\tilde{\nu}-\tilde{\nu}_i)^2}{2\sigma^2}\right), \quad (1)$$

$$l_i(\tilde{\nu}) = \frac{1}{\pi} \frac{\gamma}{(\tilde{\nu} - \tilde{\nu}_i)^2 + \gamma^2}, \quad \gamma = \sqrt{2\ln 2} \sigma \approx 1.1774 \sigma, \quad (2)$$

with σ in cm^{-1} obtained from the CLI `--sigma-ev` via $\sigma_{\text{cm}^{-1}} = \sigma_{\text{eV}} \cdot 8065.544$. Here σ is the Gaussian standard deviation ($\text{FWHM}_G = 2\sqrt{2\ln 2}\sigma \approx 2.3548\sigma$); the Lorentzian half-width-at-half-maximum γ is set to $\sqrt{2\ln 2}\sigma$ so that *both line shapes share the same FWHM* under a given `--sigma-ev` value. This removes the implicit width mismatch between the two broadening models and keeps the Gaussian–Lorentzian sensitivity comparison a pure line-shape probe; the `delta_*` columns therefore reflect tail behaviour rather than width difference. Both profiles are normalised in cm^{-1} : $\int p(\tilde{\nu}) d\tilde{\nu} = 1$. The molar extinction coefficient is then

$$\varepsilon(\lambda) = 2.315 \times 10^8 \sum_i f_i \text{profile}_i(\tilde{\nu}(\lambda)) \quad [\text{M}^{-1} \text{cm}^{-1}], \quad (3)$$

where the prefactor comes from the integrated-absorption / oscillator-strength relation $\int \varepsilon d\tilde{\nu} = 2.315 \times 10^8 \sum_i f_i$.

9.2 Beer–Lambert absorbance

$$A(\lambda) = \varepsilon(\lambda) c L, \quad \alpha(\lambda) = 1 - 10^{-A(\lambda)} \quad (4)$$

with defaults $c = 1 \times 10^{-5} \text{M}$, $L = 1 \text{cm}$ (overridable via `--concentration-m` / `--path-cm`). For experimental inputs the user-provided signal is treated as absorbance A directly and fed to the same α definition without the Beer–Lambert conversion.

9.3 Light-overlap metrics

$$\text{absorbed_flux} = \int \alpha(\lambda) I(\lambda) d\lambda, \quad (5)$$

$$\text{light_flux_total} = \int I(\lambda) d\lambda, \quad (6)$$

$$\text{absorbed_fraction} = \frac{\text{absorbed_flux}}{\text{light_flux_total}} \in [0, 1], \quad (7)$$

$$\text{shape_overlap} = \frac{\int \min(\hat{\alpha}(\lambda), \hat{I}(\lambda)) d\lambda}{\int \hat{I}(\lambda) d\lambda}, \quad (8)$$

where $\hat{\alpha}$ and $\hat{I}$ denote max-normalised shapes. `shape_overlap` is the classical geometric spectral-overlap area: at every wavelength the integrand follows whichever of the two normalised profiles is smaller, so the shaded region in a normalised overlap plot equals the numerator directly. This descriptor is independent of absolute strength and lets LED / FL sources (relative intensities) still be compared.

9.4 Choosing between `absorbed_fraction` and `shape_overlap`

The two scalar overlap descriptors answer different questions and are *both* emitted in the ranking tables and bar charts (§7.3) so that no single metric is forced on the user:

- **absorbed_fraction** is the right figure of merit when the application cares about the *number of photons (or the energy) actually absorbed* under a defined Beer–Lambert reference state (c, L) — for example, photocatalytic turnover under a given illuminant, photovoltaic action spectra, or photo-excitation yield. It folds in *both* spectral overlap and the absolute spectral intensity of the light source: a brighter source over the absorbing band is rewarded.
- **shape_overlap** is intensity-independent and answers the brightness-blind question “*how well does the light source’s spectral envelope match the molecule’s absorption envelope?*” It is the appropriate descriptor when the spectra under comparison carry incompatible intensity units

(e.g. $\text{W m}^{-2} \text{ nm}^{-1}$ for AM1.5G versus **relative** for the CIE LED templates) and the reader needs a fair side-by-side ordering.

Each descriptor is reported under both Gaussian and Lorentzian broadening (§9), so the four ranking tables together expose the full sensitivity surface to the line-shape choice.

9.5 Skip policy

An input is diverted to `skipped_inputs.*` when

- the `.out` file lacks a TD route or any **Excited State** line (tagged `INPUT_SKIPPED:fileisnotaTD-DFToutput`);
- an experimental file cannot be read or has no numeric signal column (tagged `INPUT_SKIPPED:failedtoreadexperimentalfile` or `INPUT_SKIPPED:experimentalfilehasnonnumericseries`);
- parsing fails at analysis time (tagged `ANALYSIS_ERROR:Failedtoparse...`).

10 Configuration and Environment Variables

All settings are read via `pydantic-settings` with the `GAUSS_` prefix; a `.env` file in the working directory is honoured.

Variable	Default	Purpose
<code>GAUSS_LOG_LEVEL</code>	<code>INFO</code>	Logging verbosity
<code>GAUSS_LOG_FORMAT</code>	<code>text</code>	<code>text</code> or <code>json</code>
<code>GAUSS_PLOT_OUTPUTS</code>	<code>true</code>	Emit TIFF plots
<code>GAUSS_PLOT_DPI</code>	400	Default TIFF plot resolution
<code>GAUSS_RANKING_OUTPUTS</code>	<code>true</code>	Emit grouped ranking tables and bar charts
<code>GAUSS_SIGMA_EV</code>	0.30	Default broadening σ
<code>GAUSS_WAVELENGTH_MIN_NM</code>	200.0	Grid lower bound, nm
<code>GAUSS_WAVELENGTH_MAX_NM</code>	800.0	Grid upper bound, nm
<code>GAUSS_WAVELENGTH_POINTS</code>	10000	Number of grid points
<code>GAUSS_REFERENCE_CONCENTRATION_MOLAR</code>	1e-5	Beer-Lambert c
<code>GAUSS_REFERENCE_PATH_CM</code>	1.0	Beer-Lambert L
<code>GAUSS_DEFAULT_LIGHT_SOURCES</code>	<code>AM15G,LEDB4,...</code>	Default bundled-source list
<code>GAUSS_UPLOAD_MAX_MB</code>	50	API upload size cap
<code>GAUSS_UPLOAD_ALLOWED_EXTS</code>	<code>.out,.log,.csv,.xlsx,.xls</code>	Accepted upload extensions

CLI flags override environment variables when both are supplied.

11 Troubleshooting

No output generated

Inspect `output/tables/skipped_inputs.*` first and re-run with `--log-level DEBUG`. A common cause is a `.out` file that is an OPT job, not a TD-DFT job, so it contains no **Excited State** blocks.

ParseError on an experimental file

Confirm one column is a wavelength axis (alias list in §6) and at least one other column is numeric. Non-numeric / date-formatted columns are dropped silently.

Custom light source is ignored

Check that the file has exactly one wavelength column and one numeric intensity column. The filename stem becomes the `light_source_name`; if that name collides with a bundled source the custom file wins.

CLI options seem outdated

After switching branches, re-install the package: `pip install -e ..`

Unexpected absorbed_fraction near 0 for a CIE LED source

Check that the dye's $\lambda_{\max}$ falls within the LED emission band. `shape_overlap` is often more informative than `absorbed_fraction` for narrow-band relative sources.

12 Citation and License

If you use `overlap-calculator` in published work, please cite the concept DOI below. It resolves to the latest archived release on Zenodo, so the citation stays valid across versions without needing an update.

Seyitdanlioğlu, P. *overlap-calculator: Batch spectral-overlap analysis between TD-DFT / experimental absorption and reference light sources*. Zenodo. <https://doi.org/10.5281/zenodo.19944515>

BibTeX:

```
@software{seyitdanlioglu_overlap_calculator,
  author    = {Seyitdanlioğlu, P.},
  title     = {{overlap-calculator}: Batch spectral-overlap analysis
              between TD-DFT / experimental absorption and reference
              light sources},
  year      = {2026},
  publisher = {Zenodo},
  doi       = {10.5281/zenodo.19944515},
  url       = {https://doi.org/10.5281/zenodo.19944515}
}
```

Acknowledgement of third-party data

The bundled CIE LED illuminants (LEDB1, LEDB2, LEDB3, LEDB4) and fluorescent illuminant (CIEFL10) are official **CIE** data sets derived from *CIE 015:2018 Colorimetry, 4th Edition*. The AM15G spectrum is the ASTM G-173-03 AM1.5 global reference distributed by the U.S. National Renewable Energy Laboratory (NREL). The bundled experimental absorbance workbook (`input/files/organic_uvvis_photochemcad_dataset.xlsx`) is derived from the **PhotochemCAD** spectral database. If you use `overlap-calculator` in published work, please cite the data sets alongside the software:

CIE (2018). *Relative spectral power distributions of illuminants representing typical LED lamps, 1 nm spacing*. International Commission on Illumination, Vienna, AT.
DOI: [10.25039/CIE.DS.dhcw57sd](https://doi.org/10.25039/CIE.DS.dhcw57sd)

CIE (2018). *Relative spectral power distributions of illuminants representing typical fluorescent lamps, 1 nm wavelength steps*. International Commission on Illumination, Vienna, AT. DOI: [10.25039/CIE.DS.54hy6srn](https://doi.org/10.25039/CIE.DS.54hy6srn)

CIE (2018). *CIE 015:2018 Colorimetry, 4th Edition*. International Commission on Illumination, Vienna, AT. <https://cie.co.at/publications/colorimetry-4th-edition/>

NREL. *Reference Air Mass 1.5 Spectra (ASTM G-173-03)*. U.S. National Renewable Energy Laboratory. <https://www.nrel.gov/grid/solar-resource/spectra-am1.5.html>

Taniguchi, M., & Lindsey, J. S. (2018). Database of absorption and fluorescence spectra of >300 common compounds for use in PhotochemCAD. *Photochemistry and Photobiology*, **94**(2), 290–327. DOI: [10.1111/php.12860](https://doi.org/10.1111/php.12860)

BibTeX:

```
@techreport{cie_led_illuminants_2018,
  author      = {{International Commission on Illumination}},
  title       = {Relative spectral power distributions of illuminants
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  address     = {Vienna, AT},
  year        = {2018},
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  url         = {https://doi.org/10.25039/CIE.DS.dhwc57sd}
}

@techreport{cie_fl_illuminants_2018,
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  title       = {Relative spectral power distributions of illuminants
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  year        = {2018},
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  url         = {https://doi.org/10.25039/CIE.DS.54hy6srn}
}

@techreport{cie_015_2018,
  author      = {{International Commission on Illumination}},
  title       = {{CIE} 015:2018 Colorimetry, 4th Edition},
  institution = {CIE Central Bureau},
  address     = {Vienna, AT},
  year        = {2018},
  isbn       = {978-3-902842-13-8}
}

@misc{nrel_am15g,
  author      = {{National Renewable Energy Laboratory}},
  title       = {Reference Air Mass 1.5 Spectra ({ASTM} {G-173-03})},
  howpublished = {\url{https://www.nrel.gov/grid/solar-resource/spectra-am1.5.html}},
  institution = {U.S. Department of Energy}
}

@article{taniguchi_lindsey_photochemcad_2018,
  author = {Taniguchi, Masahiko and Lindsey, Jonathan S.},
  title  = {Database of Absorption and Fluorescence Spectra of
            >300 Common Compounds for Use in {PhotochemCAD}},
  journal = {Photochemistry and Photobiology},
  volume = {94},
```



```
number = {2},
pages   = {290--327},
year    = {2018},
doi     = {10.1111/php.12860},
url     = {https://doi.org/10.1111/php.12860}
}
```

Please consult the CIE at <https://cie.co.at/> for the authoritative reference tables and the latest errata.

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